From factorial divergence to convergent coefficients

Mellin connections and inverse reconstruction in the Lieb–Liniger model

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Abstract

How can a factorially divergent expansion at one end of parameter space encode a convergent expansion at the other? We develop a coefficient-level answer, beginning with the exponential integral and ending with the repulsive Lieb–Liniger integral equation. For Stieltjes transforms, a single meromorphic moment function connects positive-index values to negative-index values, residues and finite parts. The exponential integral makes factorials, reciprocal factorials and harmonic numbers parts of one gamma-function identity. A two-parameter confluent-hypergeometric family explains Airy connection formulas and logarithmic resonance; an exactly solvable Riccati equation shows how nonlinearity and independent trans-series data modify the picture. We then prove holomorphy of the Love–Lieb resolvent on a slit parameter plane, obtain a convergent conformal expansion reaching every positive coupling, and derive the algebraic structure of Lang's regrouped strong-coupling series. First differences of the conformal coefficients give an explicit Laguerre inverse-Laplace transform. Exponentials become translated supports, allowing recursive recovery of sector germs under stated resurgent continuation assumptions. The exact transform, a finite-data smoothing bound, the conditional sector theorem and numerical evidence are kept logically separate. A recomputed 4,000-coefficient dataset detects five action thresholds and estimates the first logarithmic strength within 1.9%; it does not yet resolve higher fluctuation coefficients. The resulting framework identifies both the connection mechanism and the global information that a local asymptotic series alone can omit.

Keywords: asymptotic expansions; resurgence; Mellin transforms; coefficient interpolation; Laplace inversion; Lieb–Liniger model.

Contents

1 The connection problem and the meaning of a coefficient map 3

2 The exponential integral: one coefficient function, two endpoints 4

3 The complete Mellin dictionary, and its limitations 6

4 Airy and a master family of exact connection formulas 8

5 Riccati: nonlinearity, logarithms and two different large-parameter branches 10

6 A full-line integral equation with an exact zeta–Bernoulli bridge 12

7 Lieb–Liniger: a convergent coordinate for the full repulsive branch 13

8 Lang's regrouping: what the denominators actually encode 16

9 The weak boundary: power–logs, exponentials and physical coupling 17

10 From convergent coefficients to an exact inverse-Laplace kernel 19

11 Inverse reconstruction of trans-series sectors 21

12 Numerical tests: what is actually reconstructed 23

13 Synthesis: where the information resides 26

A Positive densities for the master family and the Riccati branch 28

B From a threshold cusp to an oscillatory coefficient tail 29

C Coefficient generation, exact test problem and reproducibility 29

1 The connection problem and the meaning of a coefficient map

A single exact function can have a divergent expansion at infinity and a convergent expansion at the origin. The disparity is especially striking when one sequence grows like $n!$ while another contains $1/n!$. This is not a contradiction: the expansions probe different parts of the same analytic object. The useful question is more demanding than whether analytic continuation connects them:

> Which explicit operations on one coefficient sequence recover the other, including logarithms, exponential sectors and their amplitudes?

There are three distinct answers, which should not be conflated. A *finite triangular map* computes a transformed coefficient from finitely many input coefficients. An *infinite-data connection* reconstructs analytic continuation or a transform from the complete sequence. A *large-order correspondence* identifies the asymptotic signature of a sector in that sequence. The first is algebraic, the second global, and the third asymptotic. None automatically provides a well-conditioned algorithm from a finite noisy prefix.

Throughout, "convergent expansion" means convergence in its stated variable. A convergent power–log expansion is not a Taylor series, and convergence in a conformal coordinate does not imply convergence of the original coupling series everywhere. In the elementary examples, $z \to \infty$ is the asymptotic endpoint. In Lieb–Liniger, weak physical coupling is the asymptotic endpoint; the auxiliary variable $B = 1/\alpha$ or w plays the role of large z.

1.1 The two mechanisms

The first mechanism is a Mellin connection. If

$$G(z) = \int_0^\infty \frac{\rho(t)}{z+t}\,\mathrm{d}t, \qquad \phi(v) = \int_0^\infty t^v \rho(t)\,\mathrm{d}t, \tag{1.1}$$

then the asymptotic coefficients of G are $(-1)^n \phi(n)$. The local coefficients at zero are obtained from the continuation of ϕ to negative indices. Regular values give powers, poles give logarithms, and finite parts fix the accompanying regular terms. This is a consequence of Mellin inversion and, in a broader class, of Ramanujan's master theorem [1, 2].

The second mechanism is an inverse transform. For a function holomorphic in a disk, its Taylor coefficients can be converted into the Laguerre coefficients of a causal inverse-Laplace kernel. The elementary identity

$$\mathcal{L}[\mathbf{1}_{t>A} g(t-A)]\,(w) = \mathrm{e}^{-Aw}\mathcal{L}[g](w) \tag{1.2}$$

turns exponentially small sectors into finite-location structure. This is classical Laplace inversion [3]; its particular application to conformal Love–Lieb coefficients supplies the inverse construction developed below.

The relation between global holomorphy and weak/strong expansions has precedents, including the analysis of finite-dimensional integrals in Ref. [4]. The present treatment does not claim a new general Mellin theorem or a new general Laguerre inversion theorem. Its purpose is to put the coefficient maps, their assumptions and their failure modes into one derivation, and to apply them to the specific integral equation.

1.2 Results and their logical status

The progression is deliberately elementary to nontrivial. Sections 2–6 give exact-function laboratories. Sections 7–9 establish the strong-origin analytic continuation and explain the weak boundary. Sections 10 and 11 give the inverse map and its conditional sector decomposition. Section 12 reports the actual finite-data tests.

Statement	What is obtained	Status
Special-function connections	Exact coefficients, logarithms and solution constants	Exact identities
Love–Lieb conformal map	A convergent strong-origin expansion for $\alpha > 0$	Proved here
Lang denominator pattern	Formal all-order regrouping and polynomial degree	Proved formally
Laguerre inverse	Full causal kernel from all conformal coefficients	Exact transform
Abel-regularized inverse	Truncation/noise bound in L^2	Proved here
All-sector recovery	Germs and their power–log coefficients	Conditional theorem
Finite-data experiment	Action locations and first cusp strength	Numerical test

Table 1: A map of the claims. The conditional step is not a proof of the full resurgent analytic class for Lieb–Liniger, and the numerical experiment is not a complete reconstruction of its trans-series.

2 The exponential integral: one coefficient function, two endpoints

2.1 Fixing the Borel convention

For an inverse-power series beginning at z^{-1}, use

$$G(z) \sim \sum_{n\geq 0} a_n z^{-n-1}, \qquad \widehat{G}(p) = \sum_{n\geq 0} \frac{a_n}{n!} p^n, \qquad G(z) = \int_0^\infty \mathrm{e}^{-zp}\widehat{G}(p)\,\mathrm{d}p. \tag{2.1}$$

The last equality requires a specified continuation and Laplace contour; it is not the definition of every function with the displayed asymptotic series. For a series beginning at z^0, the corresponding shifted sum is $z\mathcal{L}[\widehat{G}](z)$.

The alternating factorial series is represented by

$$G(z) = \mathrm{e}^z E_1(z) = \int_0^\infty \frac{\mathrm{e}^{-zp}}{1+p}\,\mathrm{d}p = \int_0^\infty \frac{\mathrm{e}^{-t}}{z+t}\,\mathrm{d}t, \qquad \Re z > 0. \tag{2.2}$$

The second representation follows by writing $(1+p)^{-1} = \int_0^\infty \mathrm{e}^{-(1+p)t}\mathrm{d}t$ and exchanging integrals. Expanding at $p = 0$ gives

$$G(z) \sim \sum_{n\geq 0} \frac{(-1)^n n!}{z^{n+1}}. \tag{2.3}$$

Thus $zG(z)$, with $x = 1/z$, realizes the original formal series $\sum(-1)^n n! x^n$. The factorial is simply the Laplace integral of p^n.

At the other endpoint, the standard convergent representation of E_1 gives [5]

$$G(z) = \sum_{n\geq 0} \frac{z^n}{n!}\big(H_n - \gamma_E - \log z\big), \qquad H_0 = 0. \tag{2.4}$$

Here a branch of $\log z$ is fixed; the two coefficient functions multiplying 1 and $\log z$ are entire. Why do the very different sequences in Equations (2.3) and (2.4) belong together?

2.2 Factorials are values; reciprocal factorials are residues

The Stieltjes density in Equation (2.2) is $\rho(t) = \mathrm{e}^{-t}$. Its moment function is

$$\phi(v) = \int_0^\infty t^v \mathrm{e}^{-t}\mathrm{d}t = \Gamma(v+1). \tag{2.5}$$

At positive indices, $\phi(n) = n!$. At $v = -n-1$, the Laurent expansion is

$$\phi(v) = \frac{(-1)^n}{n!}\left[\frac{1}{v+n+1} + H_n - \gamma_E + O(v+n+1)\right]. \tag{2.6}$$

The residues are reciprocal factorials, and the finite parts contain harmonic numbers. The Mellin identity

$$\mathcal{M}G(s) = \frac{\pi}{\sin \pi s}\phi(s-1) = \frac{\pi\Gamma(s)}{\sin \pi s}, \qquad 0 < \Re s < 1, \tag{2.7}$$

turns these Laurent coefficients directly into Equation (2.4). At $s = -n$, the two factors each have a simple pole. Multiplication by z^{-s} makes their double pole produce $-z^n \log z$.

This is the first coefficient dictionary:

$$\boxed{n! = \Gamma(n+1), \qquad \frac{(-1)^n}{n!} = \operatorname{Res}_{v=-n-1}\Gamma(v+1), \qquad \frac{(-1)^n(H_n - \gamma_E)}{n!} = \operatorname{FP}_{v=-n-1}\Gamma(v+1).} \tag{2.8}$$

There is no factorial "turning into" an inverse factorial by a termwise substitution in z. The two sequences sample different analytic data of the same function of the coefficient index.

2.3 The nonalternating series and independent exponentially small data

The companion function

$$G_+(z) = \mathrm{PV}\int_0^\infty \frac{\mathrm{e}^{-zp}}{1-p}\mathrm{d}p = \mathrm{e}^{-z}\operatorname{Ei}(z) \sim \sum_{n\geq 0}\frac{n!}{z^{n+1}} \tag{2.9}$$

has a Borel pole on the integration ray. Upper and lower contour indentations give $G_+ \pm \mathrm{i}\pi\mathrm{e}^{-z}$, respectively, with "upper" referring to a path that passes above $p = 1$. Their difference is $2\pi\mathrm{i}\mathrm{e}^{-z}$. The small-$z$ expansion is

$$G_+(z) + C\mathrm{e}^{-z} = \sum_{n\geq 0}\frac{(-1)^n z^n}{n!}\big(\gamma_E + \log z - H_n + C\big). \tag{2.10}$$

Every C gives the same algebraic asymptotic expansion at positive infinity. The equation $G_+' + G_+ = 1/z$ allows precisely this homogeneous addition. A boundary condition or a summation convention fixes it. Thus an exponentially small parameter at one endpoint becomes an ordinary local coefficient at the other.

2.4 Incomplete gamma: two sectors collide to make a logarithm

For $q > 0$, define

$$I_q(z) = \int_0^\infty \frac{\mathrm{e}^{-zp}}{(1+p)^q}\mathrm{d}p = \mathrm{e}^z z^{q-1}\Gamma(1-q, z), \qquad \phi_q(v) = \frac{\Gamma(q+v)}{\Gamma(q)}. \tag{2.11}$$

The moment representation uses $\rho_q(t) = t^{q-1}\mathrm{e}^{-t}/\Gamma(q)$, and

$$I_q(z) \sim \sum_{n\geq 0}(-1)^n (q)_n z^{-n-1}. \tag{2.12}$$

For noninteger q, the exact local decomposition is

$$\boxed{I_q(z) = \Gamma(1-q)z^{q-1}\mathrm{e}^z + \sum_{n\geq 0}\frac{z^n}{(q-1)(2-q)_n}.} \tag{2.13}$$

It follows either from the incomplete-gamma series [6] or from the residue calculation below. The two power-series factors converge for every finite z. The regular coefficients are the continued negative moments

$$(-1)^n\phi_q(-n-1) = \frac{1}{(q-1)(2-q)_n}. \tag{2.14}$$

The fractional sector comes from the other pole family.

Set $q = 1 + \varepsilon$. Then

$$\Gamma(-\varepsilon)z^{\varepsilon} = -\varepsilon^{-1} - \gamma_E - \log z + O(\varepsilon), \tag{2.15}$$

$$\frac{1}{\varepsilon(1-\varepsilon)_n} = \frac{1}{n!}\left(\varepsilon^{-1} + H_n + O(\varepsilon)\right). \tag{2.16}$$

The poles cancel between the two sectors of Equation (2.13); their finite remainder is Equation (2.4). A logarithm is therefore both a double Mellin pole and the finite remnant of colliding power-law families.

3 The complete Mellin dictionary, and its limitations

3.1 A local Borel tail is not the whole connection

Let $G = \mathcal{L}[B]$ and define $M_B(v) = \int_0^\infty p^{v-1}B(p)\mathrm{d}p$ in a nonempty fundamental strip. Exchanging the two convergent integrals gives

$$\boxed{\mathcal{M}G(s) = \Gamma(s)M_B(1-s).} \tag{3.1}$$

The continuation of this *product* governs the opposite endpoint. A formal large-p term $c\,p^{-\alpha}$ supplies the singular contribution

$$c\,\Gamma(1-\alpha)z^{\alpha-1}. \tag{3.2}$$

At integer α, the finite part introduces a logarithm. But this rule alone can omit regular powers of z.

A decisive example is

$$B_0(p) = \frac{1}{1+p}, \qquad B_1(p) = \frac{1}{1+p} + \mathrm{e}^{-p}. \tag{3.3}$$

The two functions have identical inverse-power expansions at positive infinity, while

$$\mathcal{L}[B_1](z) - \mathcal{L}[B_0](z) = \frac{1}{z+1}. \tag{3.4}$$

Every regular Taylor coefficient changes. Likewise, applying Equation (3.2) to $(1+p)^{-q}$ recovers only the first term of Equation (2.13); for $q = 3/2$ it misses the nonzero value $I_{3/2}(0) = 2$.

Proposition 3.1 (Singular terms and regularized moments)**.** *Assume that Equation* (3.1) *admits meromorphic continuation and that the relevant inverse-Mellin contour displacements are justified by bounds on vertical lines and their remainders. If M_B is regular at $n+1$, the gamma pole at $s = -n$ contributes*

$$\frac{(-1)^n}{n!}M_B(n+1)z^n. \tag{3.5}$$

If, instead,

$$M_B(v) = -\frac{c_n}{v-n-1} + h_n + O(v-n-1), \tag{3.6}$$

the contribution is

$$\frac{(-1)^n z^n}{n!}\left[h_n + c_n(H_n - \gamma_E - \log z)\right]. \tag{3.7}$$

These statements describe the residues; they do not by themselves prove convergence of the resulting local expansion.

Proof. Put $s = -n+\varepsilon$. Then $M_B(1-s) = c_n/\varepsilon + h_n + O(\varepsilon)$, while $\Gamma(s) = (-1)^n[n!\varepsilon]^{-1}[1+(H_n-\gamma_E)\varepsilon+\cdots]$ and $z^{-s} = z^n(1-\varepsilon\log z+\cdots)$. The residue gives Equation (3.7). The regular case has only the gamma pole. □

The finite parts h_n retain global information. For $B(p) = (1+p)^{-1}$, $M_B(v) = \pi/\sin\pi v$ has zero constant Laurent coefficient at its integers; this special cancellation explains why the abbreviated tail rule succeeds there.

3.2 Continuing the coefficient index

For a Stieltjes representation Equation (1.1), a direct beta integral gives

$$\mathcal{M}G(s) = \int_0^\infty \rho(t) \left[\int_0^\infty \frac{z^{s-1}}{z+t} \mathrm{d}z \right] \mathrm{d}t = \frac{\pi}{\sin \pi s} \phi(s-1). \tag{3.8}$$

The interchange is first made where absolutely convergent; subsequent use relies on continuation. The resulting inverse-Mellin integral is

$$G(z) = \frac{1}{2\pi \mathrm{i}} \int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} \frac{\pi}{\sin \pi s} \phi(s-1) z^{-s} \mathrm{d}s. \tag{3.9}$$

Proposition 3.2 (Coefficient-index dictionary)**.** *Under the contour hypotheses of Proposition 3.1, the large-z coefficients are $(-1)^n \phi(n)$. If ϕ is regular at $-n-1$, its ordinary local contribution is*

$$[z^n]_{\mathrm{reg}} G = (-1)^n \phi(-n-1). \tag{3.10}$$

If

$$\phi(v) = \frac{r_n}{v+n+1} + d_n + O(v+n+1), \tag{3.11}$$

its contribution is

$$\boxed{(-1)^n z^n (d_n - r_n \log z).} \tag{3.12}$$

Other poles of $\phi(s-1)$ yield fractional powers; higher pole multiplicities yield higher powers of $\log z$.

Proof. At $s = -n + \varepsilon$, $\pi / \sin \pi s = (-1)^n (\varepsilon^{-1} + O(\varepsilon))$. In particular this factor has no constant Laurent term. Multiplying by $\phi(s-1) z^{-s}$ proves the local formulas. The right-hand poles at $s = n+1$ give the large-z coefficients, with the reversed contour orientation accounting for the sign. □

The formula can hold without a positive density. If

$$B(p) = \sum_{n \geq 0} \frac{(-1)^n \phi(n)}{n!} p^n \tag{3.13}$$

and the analyticity and growth assumptions of Ramanujan's master theorem apply, then $M_B(s) = \Gamma(s)\phi(-s)$ and Equation (3.8) follows [2]. Positivity is a helpful sufficient structure, not a universal requirement.

3.3 Interpolation is global data, not a free substitution

Knowing $\phi(n)$ for all $n \geq 0$ does not select an arbitrary meromorphic interpolation uniquely. The exact ambiguity

$$\widetilde{\phi}(v) = \Gamma(v+1) + \frac{C}{\Gamma(-v)} \tag{3.14}$$

has the same positive-index values as $\Gamma(v+1)$, since $1/\Gamma(-n) = 0$. Nevertheless,

$$\frac{\pi}{\sin \pi s} \frac{C}{\Gamma(1-s)} = C\Gamma(s) \tag{3.15}$$

adds $C\mathrm{e}^{-z}$ to the inverse Mellin transform. Thus a term invisible at every nonnegative coefficient index can encode an entire exponentially small function. It does not satisfy the same moment/growth hypotheses as the selected Stieltjes interpolation. Nor is the added function asserted to solve the original exponential-integral differential equation.

This example makes the role of admissibility concrete. A density, a differential equation with boundary conditions, or a suitable uniqueness theorem for the interpolation fixes the missing information. Positive moment problems with a finite exponential moment are determinate: their moment generating functions are analytic near zero and are fixed by all their derivatives there. Without such a class, the formal asymptotic sequence does not determine all exponentially small additions.

3.4 The general rule for logarithms and convergence

The reliable logarithm rule is multiplicity, not integrality:

$$\boxed{\text{A pole of order } r \text{ in the full Mellin integrand produces } (\log z)^{r-1} \text{ at highest degree.}} \tag{3.16}$$

Zeros in other factors may cancel a pole. A noninteger power already accompanied by $\log p$ can produce a logarithm without any integer resonance.

Nor does Borel summability at infinity guarantee a convergent expansion at zero. For example, $B(p) = \mathrm{e}^{-\sqrt{1+p}}$ is analytic at $p = 0$ and decreases faster than every power on the positive ray, but the formal derivatives of $G = \mathcal{L}[B]$ at zero give

$$[z^n]_{\text{formal}} G = \frac{(-1)^n}{n!} \int_0^\infty p^n \mathrm{e}^{-\sqrt{1+p}} \mathrm{d}p \sim \frac{2(-1)^n \Gamma(2n+2)}{n!}. \tag{3.17}$$

The local series has zero radius. Convergence in the examples below follows from their additional analytic structure, not from the gamma kernel alone.

4 Airy and a master family of exact connection formulas

4.1 One Mellin–Barnes integral contains both expansions

For $\mu, \nu > 0$, define

$$\Phi_{\mu,\nu}(w) = w^\mu U(\mu, 1+\mu-\nu, w). \tag{4.1}$$

The standard large-argument expansion of U [7] becomes

$$\Phi_{\mu,\nu}(w) \sim \sum_{n\geq 0} \frac{(-1)^n (\mu)_n (\nu)_n}{n! w^n}, \qquad \widehat{\Phi}(p) = {}_2F_1(\mu, \nu; 1; -p). \tag{4.2}$$

The shifted Borel sum is $w\mathcal{L}[\widehat{\Phi}](w)$. The interpolation associated with Φ/w is

$$\phi_{\mu,\nu}(v) = \frac{\Gamma(\mu+v)\Gamma(\nu+v)}{\Gamma(\mu)\Gamma(\nu)\Gamma(1+v)}. \tag{4.3}$$

An equivalent and especially transparent connection is

$$\boxed{\Phi_{\mu,\nu}(w) = \frac{1}{2\pi \mathrm{i}\Gamma(\mu)\Gamma(\nu)} \int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} \Gamma(-s)\Gamma(\mu+s)\Gamma(\nu+s) w^{-s} \mathrm{d}s,} \tag{4.4}$$

where $-\min(\mu, \nu) < c < 0$ and initially $w > 0$. Moving the contour right encounters the poles of $\Gamma(-s)$ at $0, 1, 2, \ldots$ and gives the factorial expansion. Moving left encounters $s = -\mu - n$ and $s = -\nu - n$ and gives, for $\mu - \nu \notin \mathbb{Z}$,

$$\begin{aligned} \Phi_{\mu,\nu}(w) = {} & \frac{\Gamma(\nu-\mu)}{\Gamma(\nu)} w^\mu {}_1F_1(\mu; 1+\mu-\nu; w) \\ & + \frac{\Gamma(\mu-\nu)}{\Gamma(\mu)} w^\nu {}_1F_1(\nu; 1+\nu-\mu; w). \end{aligned} \tag{4.5}$$

This is Kummer's connection formula, interpreted as a coefficient map. In particular $\Phi_{1,q}(w) = wI_q(w)$, so the incomplete-gamma example is already part of this family.

The coefficient ratios make the growth contrast immediate. At infinity the ratio is

$$-\frac{(n+\mu)(n+\nu)}{n+1} \sim -n, \tag{4.6}$$

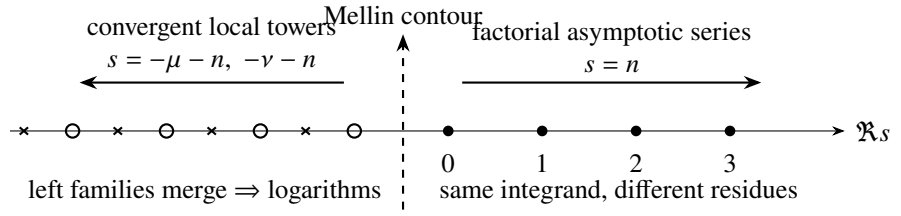


Figure 1: The Mellin–Barnes connection in Equation (4.4). Pole positions on the left are schematic. The two coefficient sequences are not matched term by term in the original variable: they arise from opposite pole families of one meromorphic integrand.

whereas in the first local tower it is

$$\frac{n+\mu}{(n+1+\mu-\nu)(n+1)} \sim \frac{1}{n}. \tag{4.7}$$

Factorial growth and inverse-factorial decay are generated by different residues of one integral, not by unrelated formulas.

When $\mu = \nu$, the two left families coalesce. Taking the limit in Equation (4.5) yields

$$\Phi_{\mu,\mu}(w) = \frac{w^{\mu}}{\Gamma(\mu)} \sum_{n\geq 0} \frac{(\mu)_n w^n}{(n!)^2} \big[-\log w + 2\psi(n+1) - \psi(\mu+n) \big]. \tag{4.8}$$

For $\mu = 1/2$, the exponent is noninteger but there is still a logarithm. This is why the pole-multiplicity rule is preferable to an "integer exponent" rule.

4.2 The Airy specialization

Set

$$\xi = \frac{2}{3}x^{3/2}, \qquad w = 2\xi, \qquad \mathcal{A}(\xi) = 2\sqrt{\pi}\, x^{1/4} \mathrm{e}^{\xi}\, \mathrm{Ai}(x). \tag{4.9}$$

The Airy–Bessel and Bessel–Kummer identities [8, 7] give

$$\mathcal{A}(\xi) = \sqrt{\frac{2\xi}{\pi}} \mathrm{e}^{\xi} K_{1/3}(\xi) = \Phi_{1/6,5/6}(2\xi). \tag{4.10}$$

Consequently,

$$\mathcal{A}(\xi) \sim \sum_{n\geq 0} \frac{(-1)^n (1/6)_n (5/6)_n}{2^n n!\, \xi^n}, \qquad B_{\mathcal{A}}(p) = {}_2F_1\left(\frac{1}{6}, \frac{5}{6}; 1; -\frac{p}{2}\right). \tag{4.11}$$

The nearest finite Borel singularity is at $p = -2$. It controls factorial large order at infinity. The large-p behavior, by contrast, contains the two power families $p^{-1/6-n}$ and $p^{-5/6-n}$; these control the opposite endpoint.

For either $\lambda = 1/6$ or $5/6$, the corresponding Borel tower is

$$D_\lambda p^{-\lambda} \sum_{n\geq 0} \frac{(-2)^n (\lambda)_n^2}{(2\lambda)_n n!}\, p^{-n}, \qquad D_\lambda = \frac{2^\lambda \Gamma(1-2\lambda)}{\Gamma(1-\lambda)^2}. \tag{4.12}$$

Using the shifted kernel $p^{-\lambda-n} \mapsto \Gamma(1-\lambda-n)\xi^{\lambda+n}$ and

$$\Gamma(1-\lambda-n) = \frac{(-1)^n \Gamma(1-\lambda)}{(\lambda)_n} \tag{4.13}$$

turns its coefficient into

$$D_\lambda \Gamma(1-\lambda) \frac{2^n (\lambda)_n}{(2\lambda)_n n!}\, \xi^{\lambda+n}. \tag{4.14}$$

One rising factorial cancels, leaving a convergent hypergeometric tower. This is an explicit coefficient-by-coefficient calculation once the Borel connection at infinity has been made.

The exact local answer can be written

$$\mathcal{A}(\xi) = C_0 \xi^{1/6} \mathrm{e}^{\xi}\, {}_0F_1\left(;\frac{2}{3};\frac{\xi^2}{4}\right) + C_1 \xi^{5/6} \mathrm{e}^{\xi}\, {}_0F_1\left(;\frac{4}{3};\frac{\xi^2}{4}\right), \tag{4.15}$$

$$C_0 = \frac{2^{-1/6}\Gamma(1/3)}{\sqrt{\pi}}, \qquad C_1 = -\frac{3\,2^{-5/6}\Gamma(2/3)}{\sqrt{\pi}}. \tag{4.16}$$

Undoing the normalization gives the entire Taylor expansion

$$\mathrm{Ai}(x) = \frac{{}_0F_1(;2/3;x^3/9)}{3^{2/3}\Gamma(2/3)} - \frac{x\,{}_0F_1(;4/3;x^3/9)}{3^{1/3}\Gamma(1/3)}. \tag{4.17}$$

The fractional powers in Equation (4.15) arise from the ramified coordinate and normalization, not from a singularity of Ai at zero.

4.3 Why the abbreviated tail rule happens to be complete here

There is no missing integer-power sector because

$$\phi_{1/6,5/6}(-n-1) = 0 \qquad (n \geq 0). \tag{4.18}$$

The numerator gamma functions in Equation (4.3) are finite at these points, while the reciprocal denominator gamma function vanishes. These are exactly the regularized negative moments that could have supplied additional ordinary powers. Their vanishing is a global property of this Borel function.

The normalized differential equation provides the same local diagnosis:

$$\mathcal{A}'' - 2\mathcal{A}' + \frac{5}{36\xi^2}\mathcal{A} = 0. \tag{4.19}$$

Its indicial roots are $1/6$ and $5/6$, with noninteger difference, so no resonant logarithm is required.

Finally, the master family has, for $\mu > 0$ and $0 < \nu < 1$, the positive density

$$\rho_{\mu,\nu}(t) = \frac{t^{\mu-1}\mathrm{e}^{-t}}{\Gamma(\mu)\Gamma(\nu)} U(1-\nu, 1+\mu-\nu, t), \qquad \frac{\Phi_{\mu,\nu}(w)}{w} = \int_0^\infty \frac{\rho_{\mu,\nu}(t)}{w+t}\mathrm{d}t. \tag{4.20}$$

For Airy this becomes the self-representation

$$\frac{\Phi_{1/6,5/6}(w)}{w} = \frac{1}{2\pi}\int_0^\infty \frac{\mathrm{e}^{-t}\Phi_{1/6,5/6}(t)}{t(w+t)}\mathrm{d}t. \tag{4.21}$$

Thus its factorial coefficients are also positive moments of a density containing the same normalized Airy function. A derivation is given in Appendix A.

5 Riccati: nonlinearity, logarithms and two different large-parameter branches

Consider

$$f'(z) + f(z) - f(z)^2 + \frac{1}{z} = 0. \tag{5.1}$$

The substitution $f = -u'/u$ yields $u'' + u' - u/z = 0$. Independent solutions are $u_1 = z$ and $u_2 = zE_1(z) - \mathrm{e}^{-z}$, so

$$f_c(z) = \frac{c + E_1(z)}{\mathrm{e}^{-z} - z(c + E_1(z))}, \qquad c \in \mathbb{C}, \tag{5.2}$$

with the exceptional solution $f = -1/z$ at $c = \infty$. This elementary linearization makes the nonlinear connection problem exact.

5.1 The factorial branch as a moment problem

The distinguished branch tending to 1 along the positive axis is $f_* = f_0$. With $G = \mathrm{e}^z E_1(z)$,

$$f_*(z) = \frac{G(z)}{1 - zG(z)} \sim 1 + \frac{1}{z} - \frac{2}{z^2} + \frac{8}{z^3} - \frac{44}{z^4} + \frac{296}{z^5} - \cdots . \tag{5.3}$$

The algebraic transformation of G is already an exact computational coefficient map. More revealingly,

$$\boxed{f_*(z) - 1 = \int_0^\infty \frac{\rho_R(t)}{z+t}\mathrm{d}t, \qquad \rho_R(t) = \frac{\mathrm{e}^{-t}}{(1 - t\mathrm{e}^{-t}\,\mathrm{Ei}(t))^2 + \pi^2 t^2 \mathrm{e}^{-2t}}.} \tag{5.4}$$

To verify it, observe that $1 - zG(z) = \int_0^\infty t\mathrm{e}^{-t}/(z+t)\mathrm{d}t$ has no zeros off the negative cut: its imaginary part has strict sign opposite to $\Im z$, and it is positive for $z > 0$. On the upper bank, $G(-t+\mathrm{i}0) = -\mathrm{e}^{-t}(\mathrm{Ei}(t) + \mathrm{i}\pi)$. Taking the imaginary part of $G/(1+tG)$ gives the density in Equation (5.4); Cauchy's formula then reconstructs $f_* - 1$. Details of the endpoint estimates are in Appendix A.

Writing

$$f_*(z) = 1 + \sum_{n\geq 0} (-1)^n \mu_n z^{-n-1}, \tag{5.5}$$

we obtain

$$\mu_n = \int_0^\infty t^n \rho_R(t)\mathrm{d}t, \qquad (\mu_0, \mu_1, \ldots) = (1, 2, 8, 44, 296, 2312, \ldots). \tag{5.6}$$

The Borel equation is particularly simple. If $h = \mathcal{B}[f_* - 1]$, then

$$(1+p)h(p) + (h * h)(p) = 1, \qquad h(p) = \int_0^\infty \mathrm{e}^{-pt}\rho_R(t)\mathrm{d}t. \tag{5.7}$$

Equating coefficients gives a positive recurrence,

$$\mu_0 = 1, \qquad \mu_n = n\mu_{n-1} + \sum_{j=0}^{n-1} \mu_j \mu_{n-1-j}. \tag{5.8}$$

Nonlinearity becomes convolution, while the spectral density makes the moment interpretation survive.

The two density endpoints explain the two expansions:

$$\rho_R(t) = t^2 \mathrm{e}^{-t}\left(1 - \frac{4}{t} + O(t^{-2})\right), \qquad t \to \infty, \tag{5.9}$$

$$\rho_R(t) = 1 + t(2\log t + 2\gamma_E - 1) + O(t^2 \log^2 t), \qquad t \downarrow 0. \tag{5.10}$$

The first yields

$$\mu_n = \Gamma(n+3)\left(1 - \frac{4}{n+2} + O(n^{-2})\right). \tag{5.11}$$

The second yields logarithms at the origin and the large-p Borel tail

$$h(p) = \frac{1}{p} + \frac{1 - 2\log p}{p^2} + O\left(\frac{\log^2 p}{p^3}\right). \tag{5.12}$$

5.2 A convergent local power–log expansion

Set $L = c - \gamma_E - \log z$ and $g(z) = \sum_{m\geq 1} (-1)^{m+1} z^m/(m\,m!)$. Then

$$f_c(z) = \frac{L + g(z)}{\mathrm{e}^{-z} - z(L + g(z))} = \sum_{n\geq 0} p_n(L) z^n, \tag{5.13}$$

where

$$p_0(L) = L, \tag{5.14}$$
$$p_1(L) = L^2 + L + 1, \tag{5.15}$$
$$p_2(L) = L^3 + 2L^2 + \tfrac{5}{2}L + \tfrac{3}{4}. \tag{5.16}$$

Substitution in the differential equation gives the polynomial recurrence

$$p'_{n+1}(L) - (n+1)p_{n+1}(L) = p_n(L) - \sum_{j=0}^{n} p_j(L)p_{n-j}(L). \tag{5.17}$$

It determines a unique polynomial of degree $n+2$. Convergence near zero on any fixed logarithmic sector follows from the exact quotient: $z\log z \to 0$, so its denominator can be expanded geometrically. Unlike the linear special functions, the nonlinear quotient can have poles away from zero. Its local coefficients need not have global inverse-factorial decay. What survives is the exact coefficient connection, not a universal rate of local coefficient decay.

5.3 The trans-series parameter belongs to the other asymptotic branch

For $c \neq 0$, put $\sigma = 1/c$ and

$$Q(z) = \frac{\mathrm{e}^{-z}}{z} - E_1(z) = \Gamma(-1, z) \sim \frac{\mathrm{e}^{-z}}{z^2} \sum_{n\geq 0} (-1)^n (n+1)! z^{-n}. \tag{5.18}$$

An exact rearrangement gives

$$\boxed{f_c(z) = -\frac{1}{z} - \frac{\sigma \mathrm{e}^{-z}}{z^2[1 - \sigma Q(z)]}.} \tag{5.19}$$

Expanding the denominator generates all sectors e^{-kz} with factorial fluctuation series. Every finite σ has the algebraic expansion $-1/z$, whereas $c = 0$ gives the distinct factorial branch $f_* \sim 1 + 1/z + \cdots$.

This distinction is important. The parameter σ is *not* an arbitrary exponentially small deformation of f_* on the positive ray. Linearization gives $\delta f' \sim \delta f$ around f_*, but $\delta f' \sim -(1+2/z)\delta f$ around $-1/z$. The decaying deformation belongs to the latter branch. At the origin the same datum is the ordinary integration constant $c = 1/\sigma$ in L.

Counterclockwise continuation around zero gives $E_1 \mapsto E_1 - 2\pi\mathrm{i}$, hence

$$c \mapsto c - 2\pi\mathrm{i}, \qquad \sigma \mapsto \frac{\sigma}{1 - 2\pi\mathrm{i}\sigma}. \tag{5.20}$$

This is an exact global monodromy relation. Splitting it into individual Stokes transformations requires the relevant sector conventions.

6 A full-line integral equation with an exact zeta–Bernoulli bridge

Before introducing interval boundaries, consider the Poisson convolution equation

$$u - P_\alpha * u = f, \qquad P_\alpha(x) = \frac{\alpha}{\pi(\alpha^2 + x^2)}, \qquad \widehat{f}(k) = |k|\mathrm{e}^{-|k|}, \quad \alpha > 0. \tag{6.1}$$

Use $\widehat{f}(k) = \int_{\mathbb{R}} \mathrm{e}^{-\mathrm{i}kx} f(x)\mathrm{d}x$. The source is $f(x) = (1-x^2)/[\pi(1+x^2)^2]$, and Fourier diagonalization gives

$$\widehat{u}(k) = \frac{|k|\mathrm{e}^{-|k|}}{1 - \mathrm{e}^{-\alpha|k|}}. \tag{6.2}$$

For $H(\alpha) = \pi u(0)$,

$$H(\alpha) = \int_0^\infty \frac{k\mathrm{e}^{-k}}{1-\mathrm{e}^{-\alpha k}}\mathrm{d}k = \sum_{m\geq 0}\frac{1}{(1+m\alpha)^2} = \alpha^{-2}\psi_1(1/\alpha). \tag{6.3}$$

The trigamma representation is standard [9]; the following connection follows directly from the integral equation.

At large α, expanding each summand for $m \geq 1$ gives the convergent series

$$H(\alpha) = 1 + \sum_{n\geq 0}(-1)^n(n+1)\zeta(n+2)\alpha^{-n-2}, \qquad |\alpha| > 1. \tag{6.4}$$

At small α,

$$H(\alpha) \sim \frac{1}{\alpha} + \frac{1}{2} + \sum_{n\geq 1} B_{2n}\alpha^{2n-1} = \frac{1}{\alpha} + \frac{1}{2} + \frac{\alpha}{6} - \frac{\alpha^3}{30} + \cdots. \tag{6.5}$$

The latter is factorially divergent. Both expansions are residues of the single Mellin transform

$$\boxed{\int_0^\infty \alpha^{s-1}[H(\alpha)-1]\mathrm{d}\alpha = \Gamma(s)\Gamma(2-s)\zeta(s), \qquad 1 < \Re s < 2.} \tag{6.6}$$

Indeed, integrate $(1+m\alpha)^{-2}$, scale by m, and sum m^{-s}. The right poles at $s = 2, 3, \ldots$ give Equation (6.4). The left poles give Equation (6.5); the functional values $\zeta(1-2n) = -B_{2n}/(2n)$ supply Bernoulli numbers, and the trivial zeros of zeta cancel the negative even gamma poles. There are no multiple poles, hence no logarithms.

This example also warns against importing a Stokes geometry merely because the kernel looks similar. Writing $z = 1/\alpha$ gives

$$H(\alpha) = z^2\int_0^\infty \mathrm{e}^{-zt}\frac{t}{1-\mathrm{e}^{-t}}\mathrm{d}t. \tag{6.7}$$

Its finite Borel singularities lie at $t = \pm 2\pi\mathrm{i}, \pm 4\pi\mathrm{i}, \ldots$. They are not the positive real actions of the finite-interval Lieb–Liniger problem. The interval boundary is an essential part of that difference.

7 Lieb–Liniger: a convergent coordinate for the full repulsive branch

7.1 Integral equation and physical normalization

The repulsive Lieb–Liniger gas has contact interaction strength $c > 0$, particle density n, and physical coupling $\gamma = c/n$. In the thermodynamic ground state, the rapidities occupy $[-Q, Q]$. Rescaling $k = Qx$ and putting $\alpha = c/Q$ gives the Love–Lieb equation [10, 11]

$$u(x;\alpha) - \frac{1}{\pi}\int_{-1}^{1}\frac{\alpha u(y;\alpha)}{\alpha^2+(x-y)^2}\mathrm{d}y = 1, \qquad -1 \leq x \leq 1. \tag{7.1}$$

We use

$$I_j(\alpha) = \int_{-1}^{1} x^j u(x;\alpha)\mathrm{d}x, \qquad \boxed{\gamma = \frac{2\pi\alpha}{I_0}, \qquad e(\gamma) = \frac{4\pi^2 I_2}{I_0^3}.} \tag{7.2}$$

Here $E_0/N = (\hbar^2 n^2/2m)e(\gamma)$ and $u(x) = 2\pi\rho(Qx)$. Strong coupling means $\alpha \to \infty$; weak coupling means $\alpha \to 0^+$. The moment problem is linear, although eliminating α to obtain $e(\gamma)$ is nonlinear.

At large γ the first terms are

$$e(\gamma) = \frac{\pi^2}{3}\left[1 - \frac{4}{\gamma} + \frac{12}{\gamma^2} + \left(-32 + \frac{32\pi^2}{15}\right)\gamma^{-3} + \cdots\right]. \tag{7.3}$$

A systematic perturbative construction and the weak-coupling trans-series are available in Refs. [12, 14]; our input below is instead the strong-origin coefficient germ.

7.2 Holomorphy of the resolvent on a slit plane

Let K_α be the integral operator in Equation (7.1) on $\mathcal{H} = L^2[-1, 1]$.

Theorem 7.1 (Slit-plane resolvent)**.** *The family $T_\alpha = (I - K_\alpha)^{-1}$ is holomorphic on*

$$\Omega = \widehat{\mathbb{C}} \setminus \mathrm{i}[-2, 2], \tag{7.4}$$

including $\alpha = \infty$. For every nonzero $g \in \mathcal{H}$,

$$\Re\langle g, T_\alpha g\rangle > 0. \tag{7.5}$$

In particular, I_0 has positive real part and never vanishes on Ω.

Proof. The kernel is continuous on the square and holomorphic in α off the slit, so K_α is a holomorphic compact-operator family. For $\Re\alpha > 0$, zero extension to the line and restriction back to the interval give

$$K_\alpha = P\mathrm{e}^{-\alpha|D|}P. \tag{7.6}$$

For each nonzero $v \in \mathcal{H}$, Plancherel's theorem implies

$$\|K_\alpha v\| \le \left\|\mathrm{e}^{-\alpha|D|}v\right\| < \|v\|\,. \tag{7.7}$$

The inequality is strict because the Fourier multiplier has modulus strictly below one except at the measure-zero set $k = 0$. A nonzero compact operator attains its norm; hence $\|K_\alpha\| < 1$. For $\Re\alpha < 0$, use $K_\alpha = -K_{-\alpha}$. For $\alpha = \mathrm{i}\beta$ with $|\beta| > 2$, the kernel is imaginary and symmetric, so K_α is skew-adjoint and $I - K_\alpha$ is invertible. Local analytic inversion proves holomorphy, and $K_\alpha \to 0$ in norm at infinity.

If $u = T_\alpha g$, taking real parts in $g = (I - K_\alpha)u$ gives

$$\Re\langle g, u\rangle = \|u\|^2 - \Re\langle u, K_\alpha u\rangle > 0. \tag{7.8}$$

Use the strict contraction off the imaginary axis and skew-adjointness on its remaining part. Choosing $g = 1$ yields the claim about I_0. □

The theorem is about the auxiliary kernel parameter. It does not assert that e is single-valued on the entire complex γ plane: the map $\alpha \mapsto \gamma$ can have critical points.

7.3 A disk containing the whole positive axis

Introduce

$$\boxed{a = \frac{\sqrt{\alpha^2 + 4} - \alpha}{2}, \qquad \alpha = \frac{1}{a} - a,} \tag{7.9}$$

with the branch $a \sim 1/\alpha$ at infinity. The inverse Joukowski map identifies Ω with $|a| < 1$; $a = 0$ is strong coupling and $a \to 1^-$ is weak coupling. It follows that

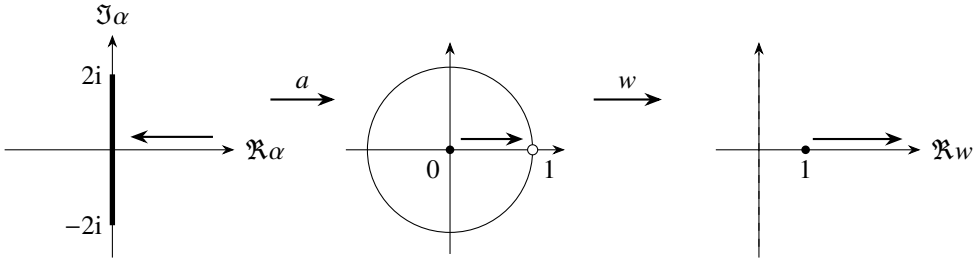
$$u(x; \alpha(a)) = \sum_{n\ge 0} v_n(x)a^n, \qquad M_j(a) := I_j(\alpha(a)) = \sum_{n\ge 0} b_n^{(j)} a^n \tag{7.10}$$

converge locally uniformly in the disk, the first in L^2. Since $I_0 \neq 0$, the energy composed with a is holomorphic there too. Every positive coupling is reached parametrically.

If the original expansion is $u = \sum_{j\ge 0} u_j \alpha^{-j}$, then

$$v_0 = u_0 = 1, \qquad \boxed{v_N = \sum_{k=0}^{\lfloor (N-1)/2\rfloor} \binom{N-k-1}{k} u_{N-2k} \quad (N \ge 1).} \tag{7.11}$$

This follows from $\alpha^{-1} = a/(1 - a^2)$. It is a finite triangular map and contains no weak-coupling input. The first terms are $v_1 = u_1$, $v_2 = u_2$, $v_3 = u_3 + u_1$ and $v_4 = u_4 + 2u_2$.



Ω: slit plane
strong → weak

$|a| < 1$: convergent germ
weak coupling at the boundary

$\Re w > 0$: Laplace variable
$w = (1 + a)/(1 - a)$

Figure 2: Three coordinates for the same analytic continuation. The physical path is $\alpha : \infty \to 0^+$, $a : 0 \to 1^-$, $w : 1 \to \infty$. The disk is a domain of ordinary convergence; its boundary carries the weak-coupling trans-series. Axes are schematic.

7.4 Uniform coefficient bounds

The positive-real-part property supplies more than holomorphy.

Corollary 7.2 (Bounded conformal moment coefficients)**.** *For each integer $j \geq 0$, the coefficients in Equation* (7.10) *obey*

$$|b_n^{(j)}| \leq B_j := 2 + \frac{2}{2j+1} \qquad (n \geq 0). \tag{7.12}$$

In particular $|b_n^{(0)}| \leq 4$ and $|b_n^{(2)}| \leq 12/5$.

Proof. For fixed real g, the function $F_g(a) = \langle g, T_{\alpha(a)} g \rangle$ has positive real part and $F_g(0) = \|g\|^2$. Its nonconstant Taylor coefficients have modulus at most $2\|g\|^2$, by integrating the nonnegative function $\Re F_g(re^{i\theta})$ against Fourier modes and sending $r \uparrow 1$. The operator is symmetric under transpose, so

$$M_j(a) = \tfrac{1}{4}\{F_{1+x^j}(a) - F_{1-x^j}(a)\}. \tag{7.13}$$

The bound is therefore $\frac{1}{2}(\|1 + x^j\|^2 + \|1 - x^j\|^2) = 2 + 2/(2j+1)$ for $n \geq 1$; the constant coefficient also satisfies it. Odd moments vanish by parity. □

These bounds will provide an explicit error estimate for a smoothed inverse transform. They do not imply that the first differences are square summable or that pointwise derivatives of the unsmoothed inverse are stable.

7.5 The energy coefficients alone are sufficient input

Let $x = 1/\gamma$ and $E(x) = e(1/x) = \sum_{n \geq 0} e_n x^n$, with $e_0 = \pi^2/3$. The exact Fermi-endpoint identity reviewed in Ref. [15, Sec. 5.3] is, in our notation,

$$\frac{\mathrm{d}B}{\mathrm{d}\gamma} = -\sqrt{\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}\gamma^2}\left(\frac{e(\gamma)}{\gamma^2}\right)}, \qquad B = \frac{Q}{c} = \frac{1}{\alpha}, \qquad B(\infty) = 0. \tag{7.14}$$

The positive root is selected by $B \sim \pi/\gamma$. In the inverse coupling,

$$B_x = \sqrt{3E + 3xE' + \tfrac{1}{2}x^2 E''}. \tag{7.15}$$

Writing $B_x = \sum s_n x^n$ gives

$$s_0 = \pi, \qquad \boxed{s_n = \frac{\frac{1}{2}(n+2)(n+3)e_n - \sum_{j=1}^{n-1} s_j s_{n-j}}{2\pi}, \qquad B(x) = \sum_{n \geq 0} \frac{s_n}{n+1} x^{n+1}.} \tag{7.16}$$

Revert the series $B(x(a)) = a/(1-a^2)$. Then

$$\boxed{M_0(a) = \frac{2\pi(1-a^2)}{a}\,x(a), \qquad M_2(a) = \frac{E(x(a))M_0(a)^3}{4\pi^2}.} \tag{7.17}$$

Each output coefficient uses only finitely many energy coefficients. For example,

$$B(x) = \pi x - 2\pi x^2 + 4\pi x^3 + \left(-8\pi + \tfrac{4}{3}\pi^3\right)x^4 + \cdots, \tag{7.18}$$

$$M_0(a) = 2 + \frac{4}{\pi}a + \frac{8}{\pi^2}a^2 + \left(\frac{4}{3\pi} + \frac{16}{\pi^3}\right)a^3 + \cdots. \tag{7.19}$$

The boundary condition in Equation (7.14) is essential: it fixes the integration constant at the convergent endpoint. The high-order computation below uses the faster kernel recurrence, while the energy-only map is checked separately at low orders.

8 Lang's regrouping: what the denominators actually encode

Lang identified a strong-coupling organization of the form [13]

$$e(\gamma) = \frac{\pi^2}{3}\frac{\gamma^2}{(\gamma+2)^2} + \sum_{n\ge 1}\frac{\pi^{2n+2}\gamma^2}{(\gamma+2)^{3n+2}}\mathcal{E}_n(\gamma), \qquad \deg \mathcal{E}_n = n-1. \tag{8.1}$$

The following derivation establishes this as an identity of formal strong-coupling series. It does not assume convergence of the infinite block sum.

8.1 Separate the constant kernel before expanding

Normalize the density by $v = u/I_0$, so $\int_{-1}^{1} v = 1$, and put $\tau = 1/\alpha$. Then $1/I_0 = \gamma\tau/(2\pi)$ and the kernel expansion gives

$$v(x) = \frac{\tau(\gamma+2)}{2\pi} + \frac{\tau}{\pi}\sum_{m\ge 1}(-\tau^2)^m (P_{2m}v)(x), \qquad P_{2m}v = \int_{-1}^{1}(x-y)^{2m}v(y)\mathrm{d}y. \tag{8.2}$$

The +2 is the constant kernel term combined with the normalization. Keeping only this term gives $v = 1/2$, $\tau = \pi/(\gamma+2)$ and exactly the leading rational block of Equation (8.1).

Set

$$r = (\gamma+2)^{-1}, \qquad q = \pi^2 r^2, \qquad \tau = \pi r T. \tag{8.3}$$

Treat q as a formal independent variable while constructing the coefficients. Then

$$v = \frac{T}{2} + r\sum_{m\ge 1}(-q)^m T^{2m+1} P_{2m}v, \qquad \int v = 1. \tag{8.4}$$

Write $T = 1 + \sum T_n(r)q^n$ and $v = 1/2 + \sum v_n(x,r)q^n$. If V_n is the order-q^n contribution of the sum, involving already known lower coefficients, then

$$T_n = -\int V_n, \qquad v_n = V_n - \tfrac{1}{2}\int V_n. \tag{8.5}$$

This is a triangular algebraic construction, not a fit to observed coefficients.

Proposition 8.1 (Formal denominator and degree theorem)**.** *The expansion Equation* (8.1) *holds formally, with rational-coefficient polynomials $\mathcal{E}_n$ of degree exactly $n-1$. Their leading coefficients are*

$$\boxed{[\gamma^{n-1}]\mathcal{E}_n(\gamma) = \frac{(-1)^{n+1}2^{2n+3}}{(n+2)(2n+1)(2n+3)}.} \tag{8.6}$$

Proof. Each nonconstant-kernel insertion in Equation (8.4) brings at least one power of q and one power of r. Induction shows that T_n and v_n are divisible by r and have degree at most n in r. Therefore

$$T^2 \int x^2 v = \frac{1}{3} + \sum_{n\geq 1} q^n C_n(r), \qquad C_n(r) = \sum_{j=1}^{n} c_{nj} r^j. \tag{8.7}$$

Since $e = \gamma^2\tau^2 \int x^2 v$, factoring the powers of r gives $\mathcal{E}_n(\gamma) = r^{-n} C_n(r)|_{r=(\gamma+2)^{-1}}$ and the stated denominator, with degree at most $n-1$. The coefficient of r comes from a single insertion. With $D_n = P_{2n}(1/2)$ it is

$$[r]C_n = (-1)^n \int_{-1}^{1} (x^2-1) D_n(x)\mathrm{d}x = \frac{(-1)^{n+1} 2^{2n+3}}{(n+2)(2n+1)(2n+3)}. \tag{8.8}$$

It is nonzero, establishing the exact degree. □

The first polynomials are

$$\mathcal{E}_1 = \frac{32}{45}, \tag{8.9}$$

$$\mathcal{E}_2 = -\frac{16(54\gamma - 53)}{945}, \tag{8.10}$$

$$\mathcal{E}_3 = \frac{128(180\gamma^2 - 306\gamma + 103)}{14175}. \tag{8.11}$$

The supplied symbolic code verifies the construction through $n = 5$.

8.2 Formal reorganization is not a global convergence theorem

Every finite sum of the displayed rational blocks is $O(\gamma^2)$ at zero, whereas the exact energy is $e(\gamma) \sim \gamma$. Any representation of the weak limit by this grouped sum must therefore involve nonuniform infinite summation. The phrase "infinite radius of convergence" used for partial resummation in Ref. [13] cannot be interpreted as an ordinary entire Taylor expansion in γ; each block has a pole at $\gamma = -2$.

There are three separate questions: convergence of the original $1/\gamma$ germ, convergence of the sum of rational blocks, and convergence of the conformal moment series. Only the third has been proved globally along the positive axis here. The normalization argument explains Lang's coefficient pattern but does not replace the slit-plane holomorphy argument.

9 The weak boundary: power–logs, exponentials and physical coupling

9.1 Where the endpoint logarithms enter

For $\alpha \to 0^+$, Equation (7.6) suggests $I - K_\alpha \sim \alpha P|D|P$, but this is not a uniform operator expansion. Boundary layers matter. The leading bulk density is a semicircle; the first correction is [11, 15]

$$u(x;\alpha) \sim \frac{\sqrt{1-x^2}}{\alpha} + \frac{x \log\frac{1-x}{1+x} + \log\frac{16\pi}{\alpha} + 1}{2\pi\sqrt{1-x^2}}. \tag{9.1}$$

This expression is nonuniform at $x = \pm 1$. Integrating its displayed terms gives, with $\ell = \log(16\pi/\alpha)$,

$$I_0 = \frac{\pi}{2\alpha} + \frac{\ell - 1}{2} + o(1), \qquad I_2 = \frac{\pi}{8\alpha} + \frac{\ell}{4} - \frac{7}{12} + o(1). \tag{9.2}$$

In physical variables,

$$\gamma = 4\alpha^2 \left[1 - \frac{\alpha}{\pi}(\ell - 1) + o(\alpha)\right], \qquad \frac{e}{\gamma} = 1 - \frac{8\alpha}{3\pi} + o(\alpha) = 1 - \frac{4}{3\pi}\sqrt{\gamma} + o(\sqrt{\gamma}). \tag{9.3}$$

The logarithms cancel in the second combination. This first cancellation is explicit; it is not an all-order no-log theorem. Logarithmic structure belongs to the observable and the parameterization, not merely to the equation.

In the disk coordinate, the same weak singularity reads

$$M_0(a) = \frac{\pi}{4(1-a)} - \frac{1}{2}\log(1-a) + O(1). \tag{9.4}$$

These two terms contribute $\pi/4$ and $1/(2n)$ to the large-order conformal coefficients. Other points of the unit circle can contribute additional terms, so this is not a complete global coefficient asymptotic by itself.

9.2 Actions and the prefactor created by changing coupling

The auxiliary-moment trans-series of Ref. [14] contains the scales

$$\mathrm{e}^{-4\pi kB}, \qquad B = 1/\alpha, \qquad k \geq 1. \tag{9.5}$$

Introduce the Cayley coordinate

$$w = \frac{1+a}{1-a}, \qquad B = \frac{a}{1-a^2} = \frac{1}{4}\left(w - \frac{1}{w}\right). \tag{9.6}$$

Then the exact action conversion is

$$\boxed{\mathrm{e}^{-4\pi kB} = \mathrm{e}^{-k\pi w}\mathrm{e}^{k\pi/w}.} \tag{9.7}$$

The first factor is a Laplace translation; the second belongs to its fluctuation series. In the disk, the same scale has a flat essential boundary behavior,

$$\mathrm{e}^{-4\pi B} \sim \mathrm{e}^{\pi}\exp\left[-\frac{2\pi}{1-a}\right]. \tag{9.8}$$

A function can therefore be holomorphic throughout the disk with an ordinary convergent Taylor series and still have a nonperturbative boundary structure.

In physical coupling, put $g = \sqrt{\gamma}/(2\pi)$. The fundamental scale is $\mathrm{e}^{-4/g} = \mathrm{e}^{-8\pi/\sqrt{\gamma}}$. However, Equation (9.2) implies

$$\boxed{\mathrm{e}^{-8\pi/\sqrt{\gamma}} \sim \mathrm{e}^2\left(\frac{\alpha}{16\pi}\right)^2\mathrm{e}^{-4\pi/\alpha} \sim \frac{\mathrm{e}^{\pi+2}}{64\pi^2}(1-a)^2\exp\left[-\frac{2\pi}{1-a}\right].} \tag{9.9}$$

The power prefactor is not optional: it comes from exponentiating the logarithm in the coupling conversion, and it changes the coefficient envelope. This is a practical reason to reconstruct the auxiliary moments first and eliminate the coupling last.

At the opposite endpoint, a fixed physical sector $\mathrm{e}^{-4k/g}$ loses its smallness as $g \to \infty$. Expanding a finite number of weak sectors at strong coupling is consequently nonuniform. The inverse construction below works with the full function rather than attempting that invalid termwise interchange.

9.3 Why the simple positive-moment interpolation does not transfer directly

Set $Z = 2\pi/\sqrt{\gamma}$ and $F(Z) = e(\gamma)/\gamma$. The first weak coefficients are

$$F(Z) \sim 1 - \frac{8}{3Z} + \frac{2(\pi^2-6)}{3Z^2} + \cdots. \tag{9.10}$$

If $F(Z)/Z$ were a positive Stieltjes transform, its first moments would satisfy $\mu_0 = 1$, $\mu_1 = 8/3$, $\mu_2 = 2(\pi^2 - 6)/3$. But the necessary moment inequality fails:

$$\mu_0\mu_2 - \mu_1^2 = \frac{2\pi^2}{3} - \frac{100}{9} < 0. \tag{9.11}$$

Signed or complex representations and other observables remain possible. What is excluded is this direct positive-density analogue of the elementary examples.

There is still a common Mellin transform,

$$\mathscr{M}(s) = \int_0^\infty Z^{s-1}F(Z)\mathrm{d}Z, \qquad -2 < \Re s < 0, \tag{9.12}$$

because $F(Z) \sim Z^2/12$ at zero and $F(Z) \to 1$ at infinity. Its continued poles would encode both endpoint expansions. But no closed gamma-product interpolation for it is derived here. In particular, an exponentially decreasing contribution integrated over $Z \geq 1$ is entire in s, so it is invisible to the large-Z pole principal parts. The global conformal germ supplies precisely the information that those principal parts alone do not.

10 From convergent coefficients to an exact inverse-Laplace kernel

10.1 A forward sector identity is not yet an inverse

A boundary exponential has an exact coefficient representation. The Laguerre generating function gives

$$\boxed{[a^n](1-a)^\beta \exp\left[-\frac{K}{1-a}\right] = \mathrm{e}^{-K} L_n^{(-\beta-1)}(K).} \tag{10.1}$$

For fixed $K > 0$, its leading large-order behavior is

$$\frac{\mathrm{e}^{-K/2}K^{\beta/2+1/4}}{\sqrt{\pi}}\, n^{-\beta/2-3/4} \cos\left(2\sqrt{Kn} + \frac{\beta\pi}{2} + \frac{\pi}{4}\right), \tag{10.2}$$

with the usual lower-order corrections [16]. Logarithmic factors are generated by differentiating with respect to β. Thus exponentially small boundary sectors can leave algebraically small, oscillatory contributions to convergent coefficients.

For a specified global function, Equation (10.1) is an exact identity. For a merely local boundary trans-series, it is a contribution to coefficient asymptotics after the required analytic hypotheses are checked. Other boundary singularities and continued lower sectors can overlap with it. Reading off frequencies is therefore not a complete inverse reconstruction.

10.2 An exact Laguerre coefficient transform

Let $M(a) = \sum_{n\geq 0} b_n a^n$ be one of the conformal moments. Put

$$H(w) = \frac{M((w-1)/(w+1))}{(w+1)^2}, \qquad q_n = \frac{b_n - b_{n-1}}{2}, \quad b_{-1} = 0. \tag{10.3}$$

The factor $(w+1)^{-2}$ smooths the inversion and makes the leading nonperturbative sector begin with w^{-2}.

Theorem 10.1 (Coefficient-to-kernel inverse)**.** *If $\{q_n\} \in \ell^2$, then*

$$\boxed{h(t) = \mathrm{e}^{-t}\sum_{n\geq 0} q_n L_n(2t), \qquad H(w) = \int_0^\infty \mathrm{e}^{-wt}h(t)\mathrm{d}t, \quad \Re w > 0,} \tag{10.4}$$

where the series converges in $L^2(0,\infty)$ *and*

$$\|h - h_N\|_2^2 = \frac{1}{2}\sum_{n>N} |q_n|^2. \tag{10.5}$$

If the q_n *have at most polynomial growth, the same formula defines a causal distribution and may be evaluated by Abel regularization,*

$$h_r(t) = \mathrm{e}^{-t}\sum_{n\geq 0} r^n q_n L_n(2t), \qquad r \uparrow 1. \tag{10.6}$$

In particular, Corollary 7.2 supplies this distributional inverse unconditionally for the Love–Lieb moments. Pointwise values and local derivatives require additional regularity.

Proof. The elementary Laplace pair is

$$\int_0^\infty \mathrm{e}^{-wt}\mathrm{e}^{-t}L_n(2t)\mathrm{d}t = \frac{(w-1)^n}{(w+1)^{n+1}} = \frac{a^n}{w+1}. \tag{10.7}$$

Also,

$$\sum_{n\geq 0} q_n a^n = \frac{1-a}{2}M(a) = \frac{M(a)}{w+1}. \tag{10.8}$$

Summing proves the transform identity. The functions $\mathrm{e}^{-t}L_n(2t)$ are orthogonal with squared norm $1/2$, proving the L^2 statements. For polynomially bounded coefficients, pairing the series with smooth rapidly decreasing test functions on the half-line is convergent: their Laguerre coefficients decrease faster than every inverse power, as follows by repeated integration by parts with the Laguerre differential operator. Extension by zero gives a causal distribution. Abel summation gives the same distributional limit, while its Laplace transform converges geometrically for $|a| < 1$. □

This is a specialization of classical Laguerre inversion [3]. Its useful feature here is the particularly simple coefficient rule: first differences of the convergent strong-origin coefficients are the Laguerre coefficients of the inverse kernel.

Unlike a perturbative Borel transform, h is constructed from the *full physical function*. It retains the information in independent one-sided contributions, not only the singularities of the perturbative germ. No weak-coupling values are sampled in constructing it.

10.3 A controlled finite-data statement

The unsmoothed inverse may be difficult, but a fixed Abel smoothing has an explicit error estimate.

Proposition 10.2 (Truncation and noise bound)**.** *Suppose* $|b_n| \leq B$ *for all n, and supplied coefficients* $\widetilde{b}_n$ *satisfy* $|\widetilde{b}_n - b_n| \leq \eta$ *for* $0 \leq n \leq N$. *Define* $\widetilde{h}_{r,N}$ *by the finite version of Equation* (10.6)*, using first differences of* $\widetilde{b}_n$. *For* $0 < r < 1$,

$$\boxed{\left\|h_r - \widetilde{h}_{r,N}\right\|_2 \leq \frac{B\,r^{N+1} + \eta}{\sqrt{2(1-r^2)}}.} \tag{10.9}$$

For M_0 *one may take* $B = 4$*, and for* M_2*,* $B = 12/5$.

Proof. The first differences satisfy $|q_n| \leq B$ and $|\widetilde{q}_n - q_n| \leq \eta$. Orthogonality gives a tail bound $Br^{N+1}/\sqrt{2(1-r^2)}$ and a finite-noise bound at most $\eta/\sqrt{2(1-r^2)}$. Add them by the triangle inequality. □

The theorem controls a smoothed global kernel, not the error in a cusp derivative or a Stokes constant. Sending $r \uparrow 1$ removes smoothing and worsens the bound. Moreover, differences between floating-point discretizations are sensitivity tests, not a proved value of η; the numerical section does not use Equation (10.9) to claim certified error bars.

11 Inverse reconstruction of trans-series sectors

11.1 State the resurgent class before asserting uniqueness

The exact transform of Theorem 10.1 is unconditional in its stated functional or distributional sense. To interpret its pieces as trans-series sectors, additional analytic structure is required.

Fix an action spacing $A > 0$ and a lateral continuation convention, denoted +. Assume that the physical inverse kernel has the locally finite representation

$$h_{\mathrm{phys}}(t) = \sum_{k\geq 0} \mathbf{1}_{t>kA}\, h_k^{+}(t - kA). \tag{11.1}$$

Each h_k is required to have a uniquely specified power–logarithmic germ at zero, to be analytically continuable along the chosen paths past subsequent thresholds, and to have growth permitting the corresponding Laplace transforms. We work in a fixed quasianalytic germ class, such as convergent power–log germs on a logarithmic cover; arbitrary additions flat at the germ origin are excluded. Distributional point contributions can be accommodated, but are not needed for the first-moment threshold analyzed below.

These are substantive assumptions. In particular, Equation (11.1) is an exact decomposition, not an equality of formal asymptotic series. The known Wiener–Hopf trans-series provides the relevant sector structure for Lieb–Liniger [14]; a general proof of all its global continuation and summability properties directly from Equation (7.1) is not supplied here.

Theorem 11.1 (Conditional triangular sector reconstruction)**.** *Within the class just specified, the complete conformal coefficient sequence, the action spacing and the lateral convention determine every sector germ. Explicitly,*

$$h_0(s) = h_{\mathrm{phys}}(s), \qquad 0 < s < A, \tag{11.2}$$

$$h_k(s) = h_{\mathrm{phys}}(kA + s) - \sum_{j=0}^{k-1} h_j^{+}((k-j)A + s), \qquad 0 < s < A. \tag{11.3}$$

Proof. The coefficient inverse determines h_{phys}. On the first interval only the zeroth sector is present, so its germ is known. Its analytic continuation along the prescribed path is unique. Suppose the first k germs and their prescribed continuations have been determined. On $(kA, (k+1)A)$, subtracting those continued terms in Equation (11.1) leaves exactly the translated kth germ. Induction gives existence of the reconstruction and uniqueness within the assumed class. □

In exact arithmetic, the continuation operation can be described by translating Taylor germs through overlapping disks along paths that avoid the singularities. In finite precision it can be severely ill-conditioned. The theorem identifies the inverse operation; it does not remove the analytic continuation problem or certify a finite-data algorithm for it.

For the auxiliary Lieb–Liniger moments, Equation (9.7) gives $A = \pi$. At the second threshold, one must subtract both the continuation of h_0 and that of h_1. A peak at 2π in a coefficient spectrum is not, by itself, the second-instanton amplitude.

11.2 From a sector germ to its fluctuation coefficients

Suppose a reconstructed germ has terms

$$h_k(s) = \sum_{m,r} d_{k,m,r}\, s^{\beta_m - 1} (\log s)^r, \qquad \Re\beta_m > 0. \tag{11.4}$$

Then

$$\mathcal{L}[s^{\beta-1}(\log s)^r](w) = \partial_\beta^r \{\Gamma(\beta) w^{-\beta}\}. \tag{11.5}$$

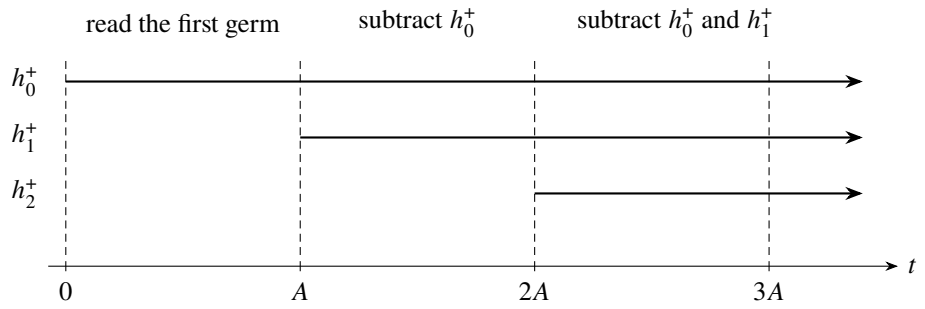


Figure 3: Exponentials become translated supports. The full kernel is the sum of the rows. Reading successive intervals only isolates sectors after the previously reconstructed germs have been analytically continued and subtracted. This is the conditional step of the inverse, not a claim that frequency fitting alone recovers every sector.

Writing the corresponding term of the fluctuation series as $\sum_\ell c_{k,m,\ell} w^{-\beta_m} (\log w)^\ell$, we obtain the finite triangular coefficient map

$$\boxed{c_{k,m,\ell} = (-1)^\ell \sum_{r \geq \ell} \binom{r}{\ell} \Gamma^{(r-\ell)}(\beta_m) d_{k,m,r}.} \tag{11.6}$$

For the first nonperturbative sector of the smoothed moment, $\beta_m = m + 2$. In the log-free case,

$$c_{k,m,0} = (m+1)! d_{k,m,0}. \tag{11.7}$$

Thus $h_1(s) = d_0 s + d_1 s^2 + d_2 s^3 + \cdots$ gives $\mathrm{e}^{-\pi w}(d_0 w^{-2} + 2d_1 w^{-3} + 6d_2 w^{-4} + \cdots)$.

A local Mellin projection makes the extraction precise. Let

$$J_k(\lambda) = \int_0^\epsilon s^{\lambda-1} h_k(s) \mathrm{d}s. \tag{11.8}$$

The term in Equation (11.4) gives the principal part

$$\frac{(-1)^r r! d_{k,m,r}}{(\lambda + \beta_m - 1)^{r+1}}. \tag{11.9}$$

Changing the cutoff changes only the regular part at this pole. The inverse procedure has therefore returned to the same residues and finite parts that governed the elementary examples, but its global input is now the conformal coefficient sequence.

For the moments, the operations from $\{b_n\}$ to sector germs are linear once the action and continuation conventions are fixed, although they are not generally finite-support operations on the sequence. Undoing $(w+1)^{-2}$ and reverting to physical coupling are subsequent nonlinear algebraic operations on the trans-series.

11.3 Stokes discontinuity versus an independent real sector

Near the first threshold, suppose the real kernel has the local form

$$h_{\mathrm{phys}}(A+s) = h_{\mathrm{reg}}(s) + C\, s \log|s| + R\, s_+ + O(s^2 \log^2|s|), \qquad s_+ = \max(s, 0). \tag{11.10}$$

The inherited perturbative singular germ is $Cs\log(-s)$. On the upper continuation, $\log(-s) = \log s - \mathrm{i}\pi$ for $s > 0$. Consequently,

$$\mathrm{Disc}\, h_0(A+s) = -2\pi \mathrm{i} C s + \cdots, \tag{11.11}$$

$$\mathrm{Disc}\, H_0(w) = -2\pi \mathrm{i} C\, \mathrm{e}^{-Aw} w^{-2} + \cdots, \tag{11.12}$$

$$h_1^+(s) = (R + \mathrm{i}\pi C) s + \cdots. \tag{11.13}$$

The logarithmic coefficient fixes the discontinuity. The real one-sided term is an additional datum of the physical completion in this convention. To quote an abstract Stokes constant rather than an observable discontinuity, the normalization of the relevant sector must also be fixed.

Orthogonality and fixed-argument Laguerre asymptotics turn Equation (11.10) into a direct coefficient signature:

$$\boxed{q_n^{(A)} \sim \frac{A^{3/4}}{2^{1/4}\sqrt{\pi}}\, n^{-5/4}\left[\pi C \sin\theta_n - R\cos\theta_n\right], \qquad \theta_n = 2\sqrt{2An} - \frac{\pi}{4}.} \tag{11.14}$$

The derivation, including the signs, is given in Appendix B. The phase is linear in $\sqrt{n}$, which permits a windowed frequency scan. Higher power–log terms generate lower-order inverse-square-root and logarithmic corrections. Neglecting them produces phase leakage between the two quadratures, a visible limitation in the numerical test.

11.4 What the complete inverse does and does not identify

Combining the maps gives

$$\begin{aligned}\{e_n\} &\xrightarrow{\text{finite recursion/reversion}} \{b_n^{(0)}, b_n^{(2)}\} \xrightarrow{\text{Laguerre inverse}} h^{(0)}_{\text{phys}}, h^{(2)}_{\text{phys}} \\ &\xrightarrow{\text{continued shifted subtraction}} \{h_k^{(0)}, h_k^{(2)}\} \xrightarrow{\text{Mellin poles / gamma derivatives}} \text{moment trans-series} \\ &\xrightarrow{\text{physical-coupling reversion}} \text{energy trans-series.}\end{aligned} \tag{11.15}$$

The full sequence determines the selected physical observable. It determines its sector amplitudes in a fixed convention, not a unique separation between an arbitrarily normalized basis sector and a trans-series parameter. If several independent sectors have the same action and local exponent structure, one scalar observable generally determines only their combined contribution unless further normalization or observable data are supplied.

Nor can finitely many unrestricted coefficients determine a nonperturbative amplitude. For any N,

$$\Delta M(a) = \varepsilon a^{N+1} \exp\left[-A\frac{1+a}{1-a}\right] \tag{11.16}$$

is holomorphic in the disk, leaves $b_0, \ldots, b_N$ unchanged, and changes the leading e^{-Aw} amplitude by ε. It need not solve the fixed Lieb–Liniger equation. Its purpose is to show why structural information or a priori bounds, not just more elaborate fits to a finite prefix, are required for controlled inverse reconstruction.

12 Numerical tests: what is actually reconstructed

The computation is designed to separate three questions: whether strong-origin coefficients reach weak coupling, whether their inverse kernel exposes the action lattice, and whether finite data determine amplitudes accurately. These are progressively harder tasks. The supplied code and data reproduce the tests below; no weak-coupling trans-series coefficients enter the generation of the conformal sequence.

12.1 Generation and independent checks

The rational kernel in the conformal coordinate gives a polynomial recurrence requiring $O(NM^2)$ operations and $O(M^2)$ working memory for N coefficients and M quadrature nodes. The recurrence is stated in Appendix C. With sufficiently many Gauss–Legendre nodes, every polynomial integral required through the retained order is exact in exact arithmetic. The implementation uses ordinary floating point, so this algebraic observation is not an interval-arithmetic certification.

A fresh run generated orders 0 through 4,000 for both moments using 2,048 nodes. Against a separately generated 2,020-node dataset, the maximum absolute changes were 2.09×10^{-10} for the I_0 coefficients and 4.78×10^{-11} for the I_2 coefficients. The changes in the corresponding first differences were 6.53×10^{-13} and 3.61×10^{-13}. The first differences, rather than the original coefficients, enter the inverse transform.

Additional tests evaluate the special-function identities at 60-decimal working precision, verify the first five Lang polynomials symbolically, and check the energy-to-moment conversion against directly generated coefficients. The suite includes an exactly solvable inverse problem with independently adjustable logarithmic and one-sided amplitudes; Appendix C.2 gives its formulas. None of these tests substitutes for a proof of finite-precision stability at all orders.

12.2 Convergent continuation to weak coupling

Table 2 compares truncated conformal moment series with direct Nyström solutions of Equation (7.1). The comparison is at fixed α: the table lists the physical γ of the reference solution. Thus it tests the parametric continuation, not a separate numerical reversion at fixed γ.

α	Reference γ	$N = 100$	$N = 200$	$N = 400$
0.50	0.5862110	1.9×10^{-11}	1.2×10^{-12}	1.2×10^{-12}
0.20	0.1213963	7.1×10^{-5}	3.3×10^{-9}	3.0×10^{-12}
0.10	0.0340254	1.2×10^{-2}	7.9×10^{-5}	3.6×10^{-9}
0.05	0.0091155	1.7×10^{-1}	1.3×10^{-2}	8.4×10^{-5}

Table 2: Relative error of $4\pi^2 I_2/I_0^3$ computed from conformal series through order N, relative to a direct 768-node solution. Repeating the direct solve with 512 nodes changes the energy by at most 1.4×10^{-13} in absolute value over these four cases. The 10^{-12} plateaux reflect the floating-point coefficient calculation and reference sensitivity; they are not bounds on the exact truncation error.

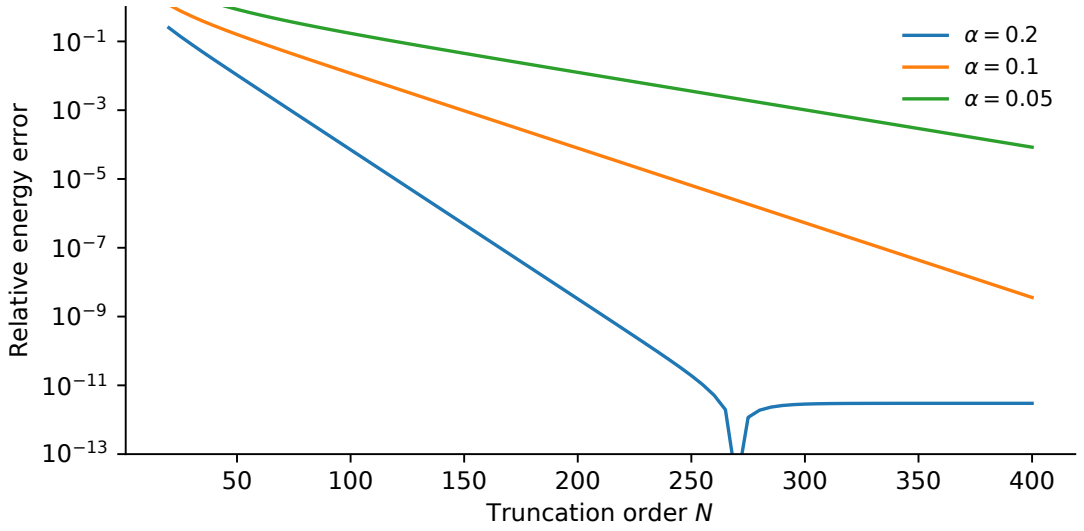


Figure 4: Convergent evaluation from strong-origin coefficients. Increasingly weak coupling moves the evaluation point toward $a = 1$, slowing convergence without changing the radius of the conformal expansion. The reference is a direct integral-equation solve, not a weak-coupling series.

The rate is consistent with

$$a = e^{-\operatorname{arsinh}(\alpha/2)} \sim e^{-\alpha/2}. \tag{12.1}$$

At fixed accuracy one consequently expects a useful coefficient count of order $1/\alpha$, up to logarithmic factors and coefficient growth. The experiment confirms access to weak coupling; it does not distinguish exponentially small sectors by subtracting two weak-coupling evaluations.

12.3 Finding the actions without imposing their locations

For the I_0 data, form $q_n = (b_n - b_{n-1})/2$. A continuous scan fits the oscillatory signature in Equation (11.14) over $0.5 < A < 16$. The fit uses the orders from approximately $0.15N$ to N, with a smooth window in $\sqrt{n}$. A constant term absorbs part of the slowly varying background. The scan locates the peaks first; the lattice $\pi\mathbb{N}$ is used only for comparison in Table 3.

Peak	Scanned action	$k\pi$	Relative displacement
1	3.13742	3.14159	−0.133%
2	6.27240	6.28319	−0.172%
3	9.40400	9.42478	−0.220%
4	12.53203	12.56637	−0.273%
5	15.65660	15.70796	−0.327%

Table 3: Action detection from 4,000 strong-origin coefficients. The systematic displacements are consistent with neglected finite-order phase corrections, not evidence for a shifted action lattice. A higher-action peak includes inherited singularities and must not be identified directly with a higher-sector amplitude.

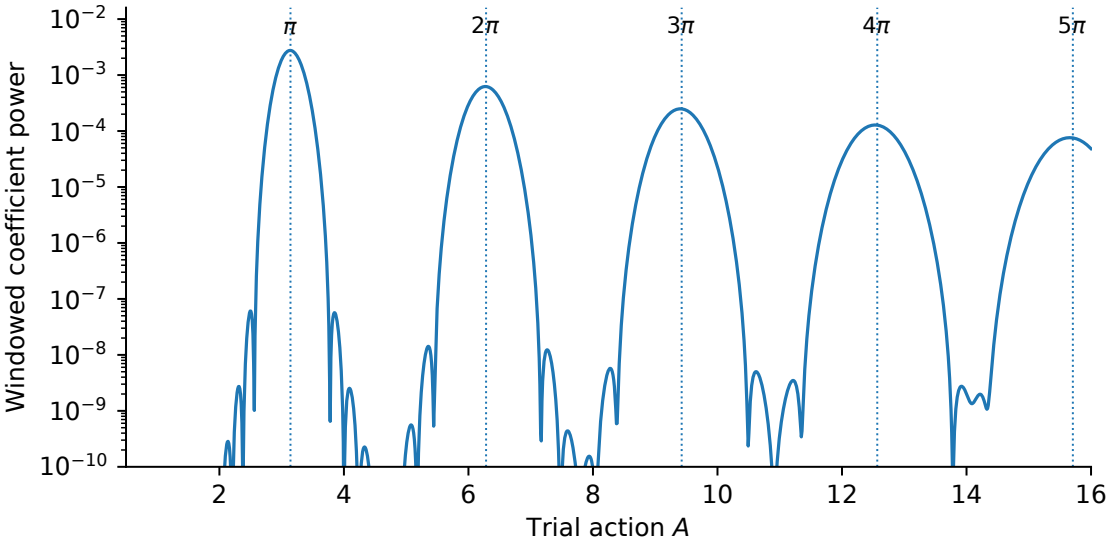


Figure 5: Windowed action scan of the conformal coefficient differences. Dashed vertical lines mark the independently known thresholds $\pi, \ldots, 5\pi$. The continuous scan, rather than a fit with those frequencies imposed, supplies the locations in Table 3. The plotted power is a detection statistic, not a trans-series coefficient.

12.4 The first logarithmic strength, and the unresolved quadrature

After detecting the first action, set $A = \pi$ and fit the leading two quadratures in Equation (11.14). In the normalization of this paper, the auxiliary-moment trans-series of Ref. [14] gives

$$I_0^{(1),+}(\alpha) \sim \frac{\mathrm{i}}{2\mathrm{e}^2}\mathrm{e}^{-4\pi/\alpha}, \qquad C_{\mathrm{ref}} = \frac{1}{2\pi\mathrm{e}^2} = 0.0215392793018486\ldots, \qquad R_{\mathrm{ref}} = 0. \tag{12.2}$$

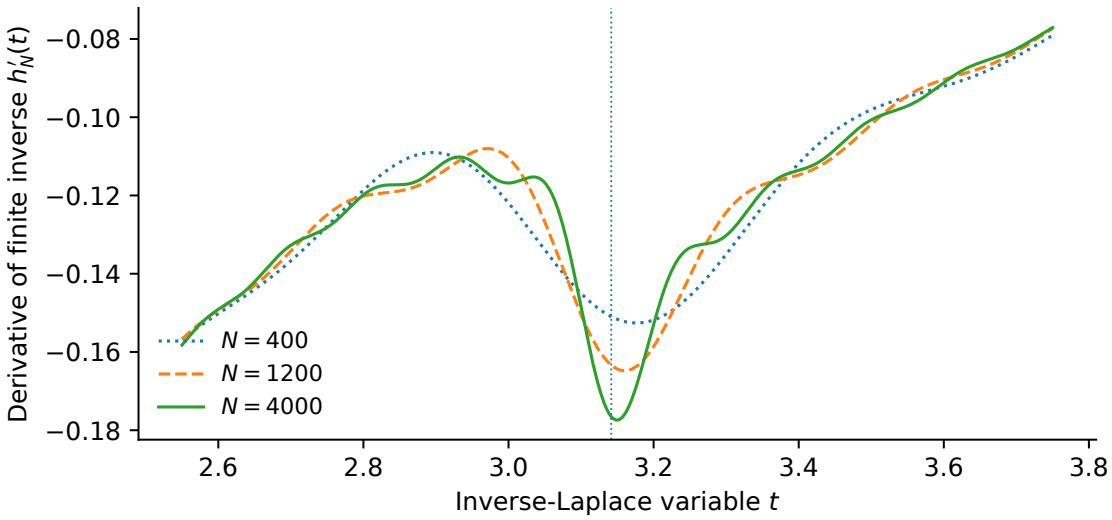


Figure 6: The derivative of the finite Laguerre inverse near the first threshold $t = \pi$. A local term $Cs \log |s|$ produces a logarithmic derivative singularity; higher truncations resolve an increasingly narrow dip. Oscillations and finite-order displacement are visible. The curves are exact derivatives of the truncated sums, not certified approximations to a singular derivative. Separating an inherited continuation from a residual sector still requires Equation (11.3).

For clarity about the normalization, that reference writes the small-gap capacitance as $C_{\text{cap}} = (\epsilon_0 \pi r^2/d)[C^{(0)} + \delta C^{(1)} e^{-2(1/\delta+1)} + \cdots]$, with $\delta = \alpha/(2\pi)$ and $C^{(1)} = 2i + O(\delta \log \delta)$. Since $C_{\text{cap}} = 2\epsilon_0 r I_0$ and $d = \alpha r$, its leading term becomes Equation (12.2). The sign changes with lateral convention. This reference value is a benchmark, not an input to coefficient generation or action detection.

Maximum order N	Fitted C	Fitted R	Relative error in C
400	0.02167675	−0.03170053	0.638%
1200	0.02201580	−0.01949256	2.21%
2000	0.02201048	−0.01551466	2.19%
4000	0.02194848	−0.01137068	1.90%

Table 4: Leading-quadrature fits at fixed $A = \pi$. The nonmonotonic error in C and substantial spurious R show that a small discrepancy at one order is not a controlled error estimate. Higher fluctuation terms have not been reliably extracted.

The numerical conclusion is deliberately limited. Five action locations are detected, and the first logarithmic/discontinuity strength is estimated at roughly the two-percent level. The real one-sided quadrature is still contaminated, higher fluctuation coefficients are unstable under changes of fitting window, and recursive subtraction through higher sectors has not been numerically validated. The formal inverse construction is stronger than this finite-data implementation.

13 Synthesis: where the information resides

The coefficient connection has two complementary forms. In the elementary examples, a meromorphic moment function places the two coefficient systems on opposite sides of the same index plane. In Lieb–Liniger, a convergent conformal germ determines a causal inverse kernel, whose translated singular structure contains the weak-coupling sectors. Both descriptions separate local singular data from the global information needed to connect endpoints.

The exponential integral gives the cleanest dictionary: values of $\Gamma(\nu + 1)$ at positive indices are

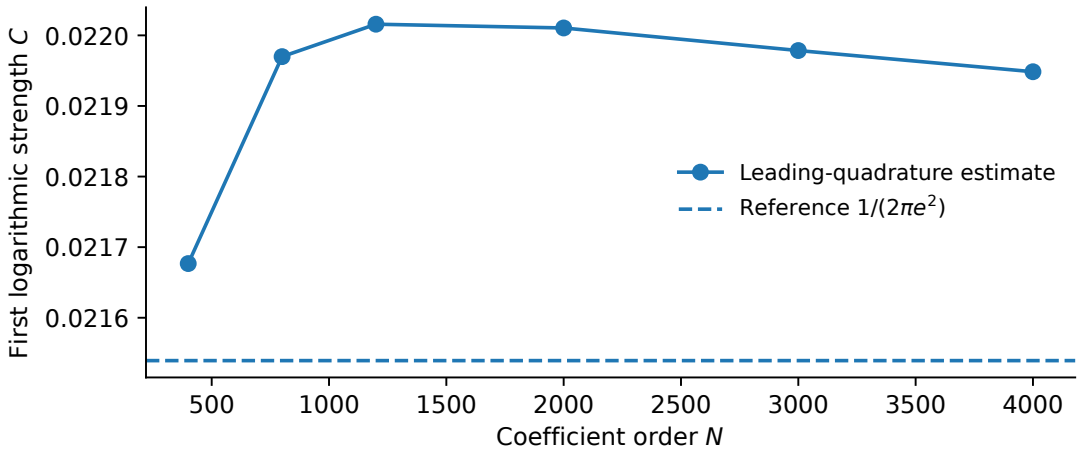


Figure 7: The fitted logarithmic strength versus the exact benchmark. The $N = 400$ value is accidentally closer than several higher-order values. This is why the quoted 1.9% is a measured discrepancy, not a certified confidence interval or a demonstrated asymptotic convergence rate.

factorials, residues at negative indices are reciprocal factorials, and finite parts are harmonic-number coefficients. The incomplete-gamma family shows why inverse-power Borel tails alone can miss an analytic sector. Airy shows how two nonresonant Mellin pole families yield two convergent local towers. Riccati shows that nonlinear transformations preserve an exact connection while changing convergence radii and introducing higher logarithmic powers; it also distinguishes the factorial branch from a different branch carrying a freely adjustable exponentially small parameter. The full-line Poisson problem demonstrates that the same Mellin mechanism extends beyond differential equations.

For finite-interval Lieb–Liniger, the endpoint is a boundary problem, not a single gamma-function identity. The analytic Fredholm argument proves a conformal domain in which the strong-origin moment series converges. The normalized kernel explains Lang's denominator and polynomial-degree pattern without proving convergence of that separate regrouping. The Laguerre transform then gives an exact inverse kernel, and its Abel-regularized version has an explicit truncation/noise estimate. Under a fixed resurgent germ class and lateral convention, translated-support subtraction recovers all sector germs and gamma derivatives convert them into fluctuation coefficients.

The remaining analytic and numerical tasks are sharply delineated. One must derive the required exact sector class and continuation bounds directly from the finite-interval equation to remove the conditional hypothesis. One must also stabilize continuation and singularity extraction to convert the infinite-data inverse into a controlled finite-data reconstruction of higher coefficients. The smoothing estimate provides one ingredient, but it does not by itself bound a singular derivative. These are specific missing estimates, not missing definitions of the inverse map.

The central lesson can be stated without conflating uniqueness with stability:

> The complete convergent coefficient sequence can encode a full trans-series, but the encoding is global. Mellin continuation or inverse-Laplace reconstruction makes the correspondence explicit; analytic class, contour convention and finite-data control determine what can actually be recovered.

A Positive densities for the master family and the Riccati branch

A.1 The hypergeometric density

For $\mu > 0$, $0 < \nu < 1$, Euler's integral gives

$$ {}_2F_1(\mu, \nu; 1; -p) = \frac{1}{\Gamma(\nu)\Gamma(1-\nu)} \int_0^1 v^{\nu-1}(1-v)^{-\nu}(1+pv)^{-\mu}\mathrm{d}v. \tag{A.1}$$

Insert

$$ (1+pv)^{-\mu} = \frac{1}{\Gamma(\mu)} \int_0^\infty t^{\mu-1}v^{-\mu}\mathrm{e}^{-t/v}\mathrm{e}^{-pt}\mathrm{d}t. \tag{A.2}$$

All factors are nonnegative for positive real variables, so exchanging the integrals is justified by Tonelli's theorem. The substitution $v = (1+y)^{-1}$ in the remaining integral produces

$$ \rho_{\mu,\nu}(t) = \frac{t^{\mu-1}\mathrm{e}^{-t}}{\Gamma(\mu)\Gamma(\nu)\Gamma(1-\nu)} \int_0^\infty y^{-\nu}(1+y)^{\mu-1}\mathrm{e}^{-ty}\mathrm{d}y. \tag{A.3}$$

The integral representation of U proves Equation (4.20). Equivalently, ρ is the density of the product of independent gamma and beta variables, with parameters μ and $(\nu, 1-\nu)$. Its Mellin moments therefore are

$$ \int_0^\infty t^v \rho_{\mu,\nu}(t)\mathrm{d}t = \frac{\Gamma(\mu+v)\Gamma(\nu+v)}{\Gamma(\mu)\Gamma(\nu)\Gamma(1+v)}, \qquad \Re v > \max(-\mu, -\nu). \tag{A.4}$$

Laplace transforming in t, and then in p, yields the Borel and Stieltjes representations. When $\nu = 1-\mu$, $U(1-\nu, 1+\mu-\nu, t) = U(\mu, 2\mu, t)$ and the reflection formula gives

$$ \rho_{\mu,1-\mu}(t) = \frac{\sin \pi\mu}{\pi} \frac{\mathrm{e}^{-t}\Phi_{\mu,1-\mu}(t)}{t}. \tag{A.5}$$

Setting $\mu = 1/6$ proves the Airy self-representation.

A.2 No omitted poles in the Riccati density

Put $D(z) = 1 - zG(z)$. The integral

$$ D(z) = \int_0^\infty \frac{t\mathrm{e}^{-t}}{z+t}\mathrm{d}t \tag{A.6}$$

has a strictly negative imaginary part when $\Im z > 0$, a strictly positive one when $\Im z < 0$, and is positive on the positive real axis. Thus D has no zero in the cut plane. The quotient $f_* = G/D$ has no off-cut poles that could add discrete terms to its dispersion representation.

For $z = -t + \mathrm{i}0$, write $G = X - \mathrm{i}Y$, with $X = -\mathrm{e}^{-t}\,\mathrm{Ei}(t)$ and $Y = \pi\mathrm{e}^{-t}$. Then

$$ \Im \frac{G}{1+tG} = \frac{\Im G}{|1+tG|^2}, \qquad -\frac{1}{\pi}\Im(f_*(-t+\mathrm{i}0) - 1) = \rho_R(t). \tag{A.7}$$

At infinity, the exponential-integral asymptotics imply $f_* - 1 = O(z^{-1})$ uniformly on the large cut-plane contours needed for the dispersion relation. At zero, $f_* = O(1 + |\log z|)$, so the small circular indentation contributes $O(r|\log r|)$ and vanishes. Cauchy's formula therefore gives Equation (5.4) with no subtraction constant and no extra residues.

The expansions Equations (5.9) and (5.10) follow by inserting the large- and small-t expansions of $\mathrm{Ei}(t)$. In particular, the density has an exponential moment for every rate smaller than one. Within positive measures with such a bound, all its moments uniquely determine its Laplace transform near zero and hence the measure. This moment uniqueness does not prohibit unrelated exponentially small additions outside that class or solutions on a different Riccati branch.

B From a threshold cusp to an oscillatory coefficient tail

Orthogonality gives

$$q_n = 2\int_0^\infty h(t)\,\mathrm{e}^{-t}L_n(2t)\mathrm{d}t \tag{B.1}$$

when $h \in L^2$, and by localization for the singular contribution considered here. For fixed $t > 0$, the Laguerre asymptotic [16] is

$$\mathrm{e}^{-t}L_n(2t) \sim \frac{\cos(2\sqrt{2nt} - \pi/4)}{\sqrt{\pi}(2nt)^{1/4}}. \tag{B.2}$$

Near $t = A + s$, linearize the phase as $\theta_n + \omega_n s$, where

$$\theta_n = 2\sqrt{2An} - \frac{\pi}{4}, \qquad \omega_n = \sqrt{\frac{2n}{A}}. \tag{B.3}$$

Localize with a smooth cutoff equal to one at $s = 0$. The leading high-frequency transforms, with positive ω, are

$$\int_{\mathbb{R}} s\log|s|\,\mathrm{e}^{\mathrm{i}\omega s}\mathrm{d}s \sim -\frac{\mathrm{i}\pi}{\omega^2}, \qquad \int_0^\infty s\,\mathrm{e}^{\mathrm{i}\omega s}\mathrm{d}s \sim -\frac{1}{\omega^2}. \tag{B.4}$$

They can be obtained by differentiating the transform of $\log|s|$ and by exponential regularization, respectively. Smooth cutoff terms do not change these leading local singular contributions. Thus the cusp part of q_n is

$$\frac{2}{\sqrt{\pi}(2nA)^{1/4}\omega_n^2}\left[\pi C\sin\theta_n - R\cos\theta_n\right], \tag{B.5}$$

which is Equation (11.14). Curvature of the phase, variation of the amplitude and higher local power–log terms give subleading corrections. Other thresholds add their own oscillations; endpoint structure at $t = 0$ supplies a separate background. The formula should therefore be used as a local asymptotic component, not as an exact representation of the complete coefficient sequence.

C Coefficient generation, exact test problem and reproducibility

C.1 A recurrence with only four history levels

Let $d = x - y$. In the conformal coordinate,

$$\frac{\alpha}{\alpha^2 + d^2} = \frac{a(1-a^2)}{1 + (d^2-2)a^2 + a^4}. \tag{C.1}$$

Define $v_0(x) = 1$, set all negative-index objects to zero, and for $n \geq 1$ let

$$R_n(x,y) = v_{n-1}(y) - v_{n-3}(y) + (2 - (x-y)^2)R_{n-2}(x,y) - R_{n-4}(x,y), \tag{C.2}$$

$$v_n(x) = \frac{1}{\pi}\int_{-1}^{1} R_n(x,y)\mathrm{d}y, \tag{C.3}$$

$$b_n^{(j)} = \int_{-1}^{1} x^j v_n(x)\mathrm{d}x. \tag{C.4}$$

These equations are the coefficient form of the integral equation, not an approximation to it. Multiplication by $a(1-a^2)$ explains why only earlier v_n enter, so no linear solve is required at each coefficient order. Evenness permits pairing positive and negative Gauss nodes. A circular buffer stores four R_n arrays and four v_n vectors.

For $n \geq 1$, the degree of v_n is at most $n - 1$. All integrands needed for $b_n^{(0)}$ and $b_n^{(2)}$ through order N consequently have degree at most $N + 1$. Gauss–Legendre quadrature with $2M - 1 \geq N + 1$ integrates these polynomials exactly in exact arithmetic. The implementation uses an even node count exceeding that requirement. Roundoff is still present and is checked by node-count changes and independent direct solves.

C.2 An exactly solvable inverse benchmark

Choose $A > 0$ and independent real constants C, R, and set

$$M(w) = -Ce^{-Aw}\,\mathrm{Ei}(Aw) + Re^{-Aw}, \qquad H(w) = \frac{M(w)}{(w+1)^2}. \tag{C.5}$$

Here $M(w)$ denotes the same function expressed in the half-plane coordinate, rather than its disk Taylor series. The principal-value inverse of $e^{-Aw}\,\mathrm{Ei}(Aw)$ is $-\mathrm{PV}(t-A)^{-1}$. Convolution with te^{-t} gives, with $s = t - A$,

$$h(t) = -C\left\{1 - e^{-t} + se^{-s}\left[\mathrm{Ei}(-A) - \mathrm{Ei}(s)\right]\right\} + R\mathbf{1}_{s>0}se^{-s}. \tag{C.6}$$

In particular, its threshold form is exactly $Cs\log|s| + Rs_+$ plus a regular term and higher corrections. This tests the normalization and sign of both quadratures independently.

For a purely coefficient-based implementation, put $e^{-Aw}\,\mathrm{Ei}(Aw) = \sum_{n\geq 0} f_n a^n$ with $w = (1+a)/(1-a)$. The first-order differential equation gives

$$(n+1)f_{n+1} + (2A - 2n)f_n + (n-1)f_{n-1} = r_n, \tag{C.7}$$

where $f_0 = e^{-A}\,\mathrm{Ei}(A)$, $f_{-1} = 0$, $r_0 = 2$, and $r_n = 4(-1)^n$ for $n \geq 1$. Also,

$$[a^n]e^{-Aw} = e^{-A}\left[L_n(2A) - L_{n-1}(2A)\right], \qquad L_{-1} = 0. \tag{C.8}$$

The same first-difference Laguerre inverse can therefore be tested without computing any numerical inverse Laplace integral.

With $A = 2.3$, $C = 0.017$ and $R = 0.011$, the 4,000-order kernel agrees with the exact expression at nine sample points away from the threshold to a maximum absolute discrepancy of 1.08×10^{-5}. The leading-quadrature fit gives $C \simeq 0.016881$ and $R \simeq 0.012804$. This deliberately imperfect recovery shows that the finite-order amplitude contamination seen in Lieb–Liniger is not, by itself, evidence against the exact inverse identity.

C.3 Contents of the reproducibility bundle

The bundle contains the manuscript source, vector figures, two conformal coefficient datasets, machine-readable test outputs, and the following scripts. `generate_coefficients.py` implements Equation (C.4); `reconstruct.py` performs the Laguerre inversion and action/strength fits; `energy_to_moments.py` checks the triangular physical-energy conversion; `lang_blocks.py` generates the exact regrouped polynomials; `verify_special_functions.py` tests the elementary identities and density formulas; `test_inverse.py` runs the exact benchmark; and `make_figures.py` regenerates the direct-solve comparisons and numerical figures.

The reported run used Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, SymPy 1.14.0 and mpmath 1.3.0. The README gives commands and dataset provenance. The code does not implement arbitrary-order stable analytic continuation of sector germs, certified Stokes extraction or a proof of the conditional hypotheses. Its reproducible scope is exactly the finite-data evidence reported above.

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